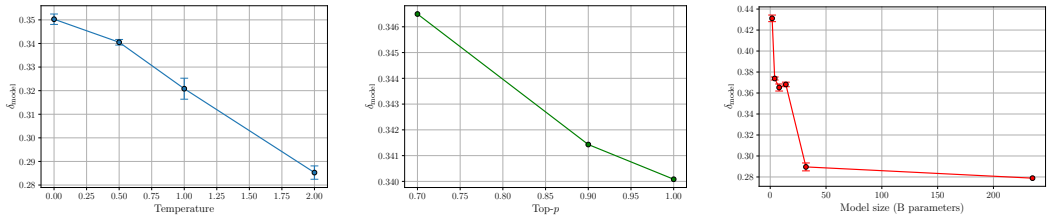

(a) Temp. (ChatGPT 3.5) (b) Top-p (ChatGPT 3.5) (c) Model size (Qwen 3)

Figure 9: Sensitivity of the combined bias metric δ_{model} to model hyperparameters. Each point is the mean over three independent runs (except for the top-p subplot); vertical bars show one standard deviation where available.

To evaluate this approach, we build a 1000-query benchmark with 8 API candidates per query and a ground truth set indicating which $K \in \{2, 3, 4, 5\}$ APIs are sufficient ($\sim$250 items each). We report subset quality: precision, recall, and exact-set match. Note that the formula for exact-set match is given by $\frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[S_i = G_i]$ where G denotes the set of ground truth sets, S the set of selected subsets, and N is the number of queries. Additionally note that bias can persist if the subset selector itself is biased; underselecting viable tools (false negatives) or selecting unrelevant ones (false positives). Measures of recall and precision will tell us whether this is the case.

E More Elaborate Ablation and Sensitivity Analysis

Temperature. Raising temperature reduces combined bias. As shown in Figure 9a, as temperature goes from 0 to 2, the mean δ_{model} for ChatGPT 3.5 drops from about 0.350 to 0.285, a 6.5% absolute reduction. Figure 10 makes clear why: the overall selection patterns remain similar across temperatures, but higher temperatures soften extreme preferences. This suggests that increased stochasticity slightly mitigates bias, but does not eliminate it.

Top-p. Figure 9b shows how the combined bias δ_{model} for ChatGPT 3.5 varies with the top-p cutoff. Increasing top-p from 0.7 to 1.0 yields a small decrease in bias (from $\sim$0.346 to $\sim$0.340), suggesting that less aggressive truncation of the probability distribution slightly softens extreme tool preferences. The effect is noticeably weaker than the temperature change.

Model Size. In Figure 9c, the combined bias δ_{model} is depicted for Qwen 3 with varying model size. It seems that larger models exhibit less bias, with a notable drop at 32B. This pattern suggests that larger models develop more nuanced selection mechanisms which temper extreme preferences for certain APIs.

API Ordering. Figure 11 compares ChatGPT 3.5's API selection under two different ordering schemes: cyclic rotations versus random permutations. Across all clusters, the choice distribution is very similar: no API's selection rate shifts more than about ten percentage points. This indicates that the ordering of the APIs has some influence, but the dominant signal is the model's intrinsic preference. However, the small differences could also reflect the inherent noise from stochastic token sampling, and overall we argue that the tool-selection behavior is robust to either type of shuffling.

System Prompts. To evaluate how sensitive tool selection is to the phrasing and structure of the instructions given, we compare three variants of the system prompt: the original "Base" prompt, a lightly reworded "Similar" prompt, and a structurally different "Adjusted" prompt. Figure 12 shows the resulting distributions for ChatGPT 3.5.

Prompt wording shifts model preferences but does not remove bias. Reworded prompts can amplify dominant choices and in some cases radically redistribute the selection shares. Elsewhere, effects are modest. Overall, framing and formatting can tilt the implicit ranking among functionally equivalent

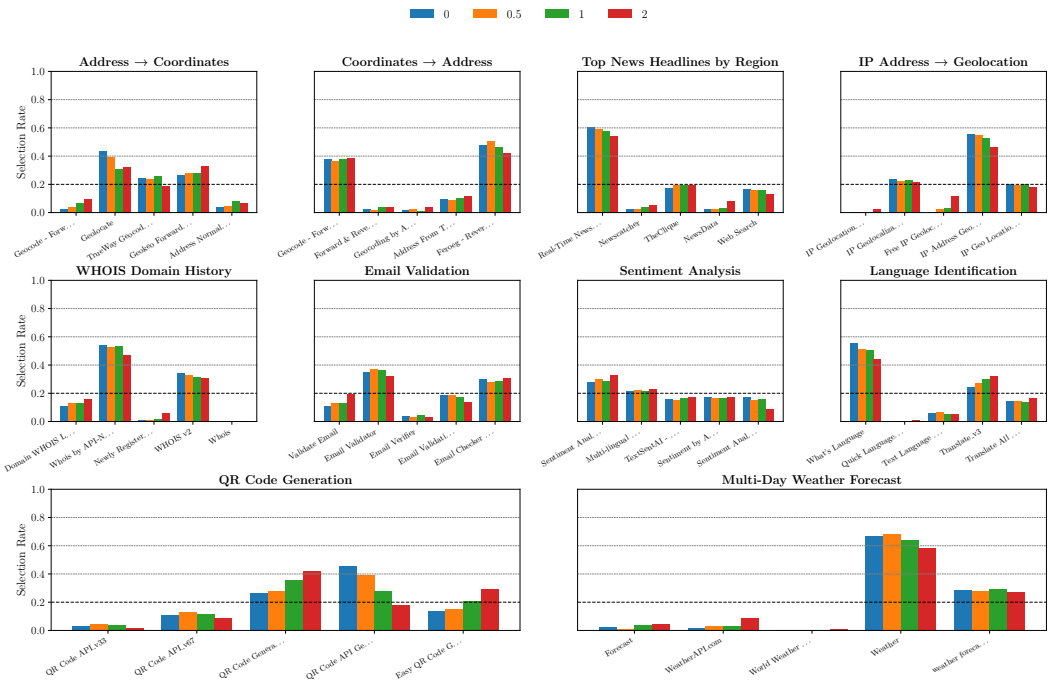

Figure 10: Selection distributions for ChatGPT 3.5 using four different temperatures across ten clusters of functionally equivalent APIs. Each subplot corresponds to one cluster, with the x-axis indicating the API in the cluster and the y-axis showing the fraction of times that API was chosen by the respective model over 500 runs.

APIs, indicating that part of the observed bias is prompt-dependent even as a tendency to favor a subset of tools remains.

F More Elaborate Discussion on the Explanation of Bias

We now expand on the analysis given in the main text surrounding the investigation of bias. We expand on the feature-level analysis, where we try to predict selection rates according to intrinsic API attributes, and on the perturbation experiments that directly intervene on the API metadata to see which cues the models rely on during selection.

F.1 Which API-level features predict selection rates?

We extract a common set of descriptive features from every API (see Section 3.3) and mean-center them to investigate how being relatively high or low on a feature affects the API selection. These are then paired with the empirical selection rates yielding a dataset of 50 examples for each LLM. We then probe relationships between features and selection behavior in three ways. First, we compute Pearson correlations to capture linear and monotonic associations. Second, we fit a linear regression per model to quantify the aggregate explanatory power (reported as R^2) and inspect coefficients to understand the direction and relative weight of each feature. Third, we train random-forest regressors with cross-validation to allow for non-linear interactions and obtain alternative measures of feature importance.

Similarity between tool / API description and query is most correlated to selection rate. As Table 4 makes clear, the most predictive feature of API selection is semantic similarity between the query and the tool/API descriptions. Both `avg_similarity_tool_desc` and `avg_similarity_api_desc` are consistently positively correlated with selection rates—especially strong for Qwen and clear for Gemini; ChatGPT shows the same pattern, albeit

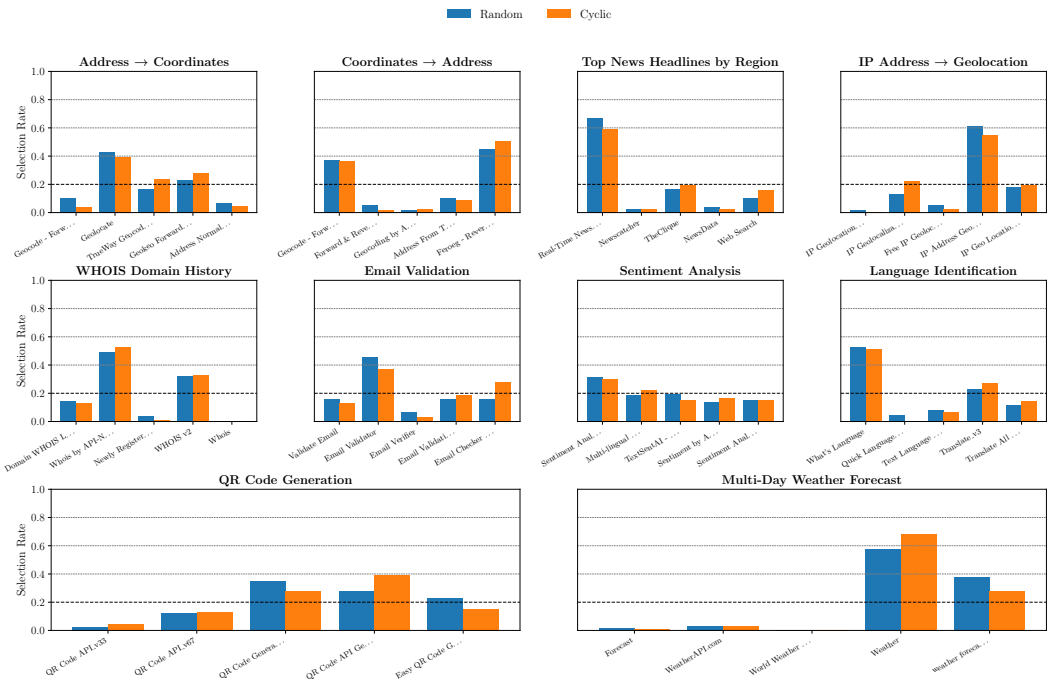

Figure 11: Selection distributions for ChatGPT 3.5 using cyclic and random shuffling of the APIs across ten representative clusters. Each subplot corresponds to one cluster, with the x-axis indicating the API in the cluster and the y-axis showing the fraction of times that API was chosen for that specific API ordering method over 500 runs.

Table 4: Correlation between API-level features and model selection rates. Each entry shows Pearson r with its p-value.

Feature	ChatGPT 4.1	Gemini	Qwen
`avg_similarity_tool_desc`	$+0.227$(p=0.113)	-0.092(p=0.526)	$+0.234$(p=0.101)
`avg_similarity_api_desc`	$+0.111$(p=0.442)	$+0.330$(p=0.019)	$+0.411$(p=0.003)
`age_days`	-0.199(p=0.201)	-0.144(p=0.356)	-0.163(p=0.296)
`desc_name_length_sum`	$+0.044$(p=0.760)	$+0.103$(p=0.477)	$+0.038$(p=0.795)
`num_params`	-0.065(p=0.653)	$+0.038$(p=0.793)	-0.185(p=0.198)
`flesch_reading_ease`	$+0.160$(p=0.267)	$+0.176$(p=0.222)	$+0.098$(p=0.496)
`positive_word_count`	$+0.126$(p=0.384)	$+0.087$(p=0.547)	$+0.093$(p=0.521)

weaker with higher p-values. By contrast, structural or stylistic attributes (e.g., parameter count, promotional wording) exhibit little consistent influence. Tool age (`age_days`) shows a modest, broadly consistent negative correlation across models.

Linear regression reveals that surface-level semantic alignment is the primary signal but leaves a lot unexplained. Figure 13 tells us that linear models explain only part of the variance: R^2 is modest—0.143 for ChatGPT-4.1 and 0.387 for ToolLLaMA—leaving substantial error. Coefficients show surface-level semantic alignment dominates: similarity between the query and tool/API descriptions has the largest positive weights for most models, with Qwen weighting both most strongly and Gemini emphasizing API-level descriptions. Unexpectedly, ToolLLaMA gives a negative weight to tool-description similarity. Other features contribute little. Hence, semantic alignment is important in driving selection but still gives an incomplete explanation, implying nonlinear or omitted factors and motivating more flexible models (e.g., random forests).